

Raw Document NME:

### NME – Non Medical Expenses

The following items will not be covered by the insurance & corresponding amount must be collected from the patient:

1. Instrument Charges (Only GIPSA it is an NME. For other insurance companies, even if it is billed we do not collect from the patient) – DNB
2. Laser Equipment
3. Disposables:
  - Gloves (If included in pkg, do not collect)
  - Masks (Nebulization/Oxygen/Cpap) (If included in pkg, do not collect)
  - Underpads
  - EZ Bath
  - Ortho Roll (If included in pkg, do not collect)
  - Dressing Pads (If included in pkg, do not collect)
  - HME Filters (If included in pkg, do not collect)
  - Hand Rubs (If included in pkg, do not collect)
  - Caps (If included in pkg, do not collect)
  - Splints (If included in pkg, do not collect)
  - Drapes (If included in pkg, do not collect)
  - Tegaderm (If included in pkg, do not collect)
  - Dettol
  - Thermometer
  - Mouthwash
1. Equipment:
  - Alpha Bed (Alphaexcel etc)
  - Warmers
1. Hospital Services:
  - Patient Gowns
  - Linen
  - Laundry
  - FnB
  - Dental Care
  - Medical Record Charges
  - MLC Charges
  - Diet Consultations
  - Ambulance Charges
  - Additional CD/DVD charges
1. Investigations & Procedures:
  - Blood Grouping (If included in Pkg, do not collect)
  - Viral Screening-HIV/HCV/HBsAG (If included in Pkg, do not collect)

- TSH/T3/T4, for non-diabetic patients or for those cases with no relevance to the clinical condition for which the patient is admitted
- Vitamins (D, C, A, E, B12)
- Iron – If they are normal, then they are not paid by the Insurance company
- Any investigations prescribed by Cross consultations (if results are normal; if the values are abnormal & there is corresponding treatment for that, the test & treatment are covered)
- GRBS is not covered if Diabetic chart is not sent earlier

#### **Package NMEs:**

1. Medical records
2. GRBS (Case by case basis)
3. EZ bath
4. Under pads
5. F & B
6. Ward consumables
7. Splints, binders, walkers

#### **Codex Ouptut**

The best approach is a hybrid NME estimator:

Policy rules decide whether an item is patient-payable; historical HIMs NME amounts decide the likely range; known treatment-specific items adjust that range.

A flat percentage of the total bill or a universal ₹5,000/₹10,000 allowance will not work well.

I reviewed the complete [hospital NME notes](/Users/reylvanttambi/Downloads/Reylvant - General Doc 6 (7\).docx) and validated the available NME fields and line items against the clean historical IP data.

## **What data we already have**

Out of 13,974 clean admissions:

- 13,843 have a final **HIMs NME Amount**.
- 4,212 have a positive NME amount.
- 7,772 have an NME amount entered by the financial counsellor.

- We have treatment, payer, package/open status, LOS, ICU days, room category and service/pharmacy lines for modelling.

The current counsellor estimates are commonly ₹5,000 for open bills and ₹10,000 for packages. They perform poorly:

- Only approximately 11–24% of open-insurance estimates were within  $\pm 25\%$  of actual NME.
- Only 0.3% of GIPSA package estimates and 2% of Non-GIPSA package estimates were within  $\pm 25\%$ .

These fixed defaults should be replaced.

## Policy interpretation

### Cash

NME is an insurance non-payable concept. For Cash:

- Do not show a separate “NME payable” amount.
- Include the relevant consumables, equipment and services in the ordinary Cash estimate.

### Insurance

Every charge must resolve to one of:

1. Covered by insurer.
2. Patient-payable NME.
3. Do Not Bill—neither insurer nor patient.
4. Conditional/unknown at estimate time.

This classification must happen before estimating the NME amount.

### Instrument charges

- GIPSA: instrument charges are patient-payable NME.
- Non-GIPSA: instrument charges may remain in the insurer bill, but should not be collected from the patient; if denied, they become DNB.
- Corporate: organization-specific.
- Cash: part of the normal Cash estimate.

This makes payer-specific logic essential.

### Package precedence

For items marked “If included in package, do not collect”:

- Check the selected package’s inclusion/exclusion rules.
- A package-specific inclusion overrides the general NME list.
- Do not collect an included item separately.
- If package documentation is unresolved, show the amount as a conditional NME risk.

This applies to gloves, masks, ortho roll, dressing pads, HME filters, hand rub, caps, splints, drapes, Tegaderm, blood grouping and viral screening.

## Historical NME behaviour

### Overall insurance results

| Payer      | Admissions with actual NME | Positive NME | Positive rate | Positive NME P25–P75 |
|------------|----------------------------|--------------|---------------|----------------------|
| GIPSA      | 3,307                      | 1,088        | 32.9%         | ₹1,753–₹8,978        |
| Non-GIPS A | 4,702                      | 3,101        | 66.0%         | ₹3,373–₹9,383        |

These totals are too broad to use directly because package/open status and LOS materially change NME.

### Open medical management

Typical positive-NME ranges:

| LOS               | GIPSA           | Non-GIPSA       |
|-------------------|-----------------|-----------------|
| 0–1 day           | ₹1,200–₹2,000   | ₹1,400–₹2,200   |
| 2–3 days          | ₹3,400–₹6,000   | ₹3,400–₹5,900   |
| 4–5 days          | ₹4,700–₹9,000   | ₹4,700–₹9,600   |
| 6–7 days          | ₹8,200–₹16,900  | ₹6,900–₹15,800  |
| 8–10 days         | ₹12,500–₹22,200 | ₹11,500–₹25,900 |
| More than 10 days | ₹22,100–₹44,600 | ₹15,500–₹55,600 |

## Open procedural treatment

| LOS       | GIPSA           | Non-GIPSA       |
|-----------|-----------------|-----------------|
| 0–1 day   | ₹3,000–₹6,000   | ₹2,400–₹5,800   |
| 2–3 days  | ₹5,100–₹9,300   | ₹5,400–₹9,100   |
| 4–5 days  | ₹9,500–₹15,600  | ₹9,500–₹13,900  |
| 6–7 days  | ₹11,800–₹17,600 | ₹9,600–₹20,200  |
| 8–10 days | ₹16,900–₹33,900 | ₹18,400–₹35,000 |

The aggregated 11-plus-day procedural cohort is clinically mixed and should not be used as a generic lookup. It requires exact treatment or treatment-family history.

## Package cases

Package NME behaves very differently:

| Payer             | Positive-NME frequency | Positive NME median |
|-------------------|------------------------|---------------------|
| GIPSA package     | 10.8%                  | ₹280                |
| Non-GIPSA package | 27.5%                  | ₹5,450              |

However, these generic figures should not be used when a known major instrument, laser, splint or other treatment-specific NME is expected.

The package-specific history can differ substantially:

- GIPSA package TKR frequently has no recorded NME or only a medical-record-type amount.
- Non-GIPSA package TKR positive cases often show approximately ₹8,000–₹12,000.
- Package CAG frequently has no NME, but positive cases range from approximately ₹1,500–₹6,000.
- Package laparoscopic appendectomy/cholecystectomy positive cases commonly fall around ₹4,000–₹8,000.

This supports exact treatment/package profiles rather than one universal package allowance.

## Recommended estimator

The calculation should have three components:

Estimated NME

= routine NME subtotal

+ planned treatment-specific NME

+ conditional historical reserve

## 1. Routine NME subtotal

Calculate predictable recurring items from LOS and package rules.

Historically observed insurance rates are approximately:

| Item                              | Code                  | Typical rate                        |
|-----------------------------------|-----------------------|-------------------------------------|
| Ward consumables                  | HSP5013               | ₹350 per eligible day               |
| Medical records—one day           | MSC10                 | ₹600                                |
| Medical records—more than one day | RNS0120               | ₹1,500                              |
| Diet consultation                 | DIE0001               | ₹740–₹770                           |
| F&B                               | Package-bill evidence | Approximately ₹35–₹155 per day      |
| GRBS                              | EME0020               | ₹220–₹240 per instance              |
| Alpha bed                         | EME0021               | Approximately ₹590 per day          |
| Warmer                            | EME0087               | Approximately ₹930 per instance/day |

These are historical observations. Runtime must select current tariff rates.

Example for a three-day admission:

|                            |               |
|----------------------------|---------------|
| Medical records            | ₹1,500        |
| Ward consumables: 3 × ₹350 | ₹1,050        |
| Diet consultation          | ₹750          |
| F&B: 3 × ₹35–₹155          | ₹105–₹465     |
| Routine subtotal           | ₹3,405–₹3,765 |

After ordinary disposables and conditional investigations, the observed total for a three-day open procedure is usually approximately ₹5,000–₹10,000.

## 2. Planned treatment-specific NME

Link predictable NME items to the clinical treatment master.

Examples:

| Treatment feature                         | Expected NME handling                                          |
|-------------------------------------------|----------------------------------------------------------------|
| Major surgery using OT instruments        | Instrument NME for GIPSA only                                  |
| Urology laser treatment                   | Laser charge for applicable insurance routes                   |
| Orthopaedic procedure                     | Splint, binder, walker or ortho roll where clinically expected |
| Arthroscopy                               | Arthroscopy instrument charge for GIPSA                        |
| Prolonged/immobile stay                   | Alpha bed probability                                          |
| Neonatal/temperature-management treatment | Warmer                                                         |
| Respiratory/ICU treatment                 | Masks, HME filters, underpads and related disposables          |

Historical service rates illustrate the potential size:

- Major instrument charge: approximately ₹8,850 for GIPSA.
- Arthroscopy instrument charge: approximately ₹12,500 for GIPSA.
- Laser equipment: approximately ₹37,600 median for GIPSA and ₹14,400 for Non-GIPSA.
- Medium/minor instrument charges are lower but tariff-dependent.

These large planned items must be modelled separately because they can dominate the NME range.

## 3. Conditional historical reserve

Some NME cannot be known during initial FC:

- Vitamins.
- Iron investigations where results are normal.
- Thyroid tests without treatment relevance.
- Cross-consultation investigations that return normal.
- GRBS without adequate diabetes documentation.
- Unexpected disposables.

- Ambulance, MLC, dental or additional media charges.

For each treatment/context, calculate:

residual NME =

final HIMS NME

- recognized routine NME
- recognized planned treatment NME

Use the historical residual distribution as the conditional reserve. This avoids counting the same major instrument or daily consumable twice.

## Historical profile selection

The Builder should search for the most specific reliable profile in this order:

1. Exact treatment/combo + organization + package + LOS band + ICU context.
2. Exact treatment/combo + payer bucket + package/open + LOS band.
3. Treatment family + payer bucket + package/open + LOS band.
4. Department + procedure/medical route + payer + package/open + LOS band.
5. Global payer + procedure/medical route + package/open + LOS band.

Minimum recommended sample:

- At least 30 admissions for an automatic profile.
- Between 15 and 29: use the profile but blend it with the next fallback level.
- Fewer than 15: use the fallback profile and mark confidence low.

Use recent effective history, preferably the latest 18–24 months.

## Producing the patient-facing range

NME is a zero-inflated outcome: many admissions have no NME, while a smaller subset can have substantial NME.

For each profile calculate:

- Probability of positive NME.
- P25, P50, P75 and P90 among positive cases.
- Sample size.
- Historical recency.
- Known deterministic add-ons.

Recommended display logic:



## **Positive probability of at least 70%**

Display:

Estimated non-medical expenses: ₹P25–₹P80

## **Positive probability of 30–69%**

Display:

Estimated non-medical expenses: ₹0–₹P80

NME may apply depending on package inclusions and final insurer review.

## **Positive probability below 30%**

Display:

Possible non-medical expenses: up to ₹P75

If a known mandatory item such as laser equipment or a GIPSA instrument charge exists, add that item explicitly even when the overall historical positive probability is low.

Round patient-facing values to the nearest ₹500 or ₹1,000. Do not display ranges such as ₹5,431–₹9,128.

# **Package and open-bill handling**

## **Open insurance bill**

- Identify all NME lines.
- Separate them from the insurer-payable bill.
- Remove DNB items.
- Present the patient-payable NME range separately.

## **Hospital package**

- Check package-specific inclusions and exclusions.
- Remove included items from patient NME.
- Add package NMEs and excluded treatment-specific NME.
- Do not rely on the generic package list if the selected package agreement says otherwise.

## **GIPSA package**

- Use the GIPSA agreement/package inclusions.
- Treat applicable instrument charges as NME.
- Calculate package NME separately from package exclusions payable by the insurer.
- Do not classify every package exclusion as patient-payable NME.

## Non-GIPSA package

- Do not collect instrument charges from the patient.
- Include other patient-payable NME according to the MOU and package rules.
- If an instrument charge is denied, classify it as DNB rather than NME.

## Conditional investigations

The Builder should not attempt to predict actual lab results.

Use inputs where known:

- `is_diabetic.`
- `diabetic_chart_available.`
- `investigation_clinically_related.`
- `cross_consultation_expected.`
- `package_includes_preoperative_screening.`

At initial estimate time:

- Clearly expected and uncovered investigation: add deterministically.
- Clearly covered: exclude from NME.
- Result-dependent: place in conditional reserve.
- Unknown: do not silently add as certain patient liability.

## Multiple treatments

Do not add the complete NME range of every treatment; that would double-count recurring items.

For combinations:

1. Calculate medical records once.
2. Calculate ward consumables and F&B once per stay day.
3. Union and deduplicate planned disposables/equipment.
4. Add treatment-specific instruments/laser separately.
5. Use direct historical combo data when sample size is sufficient.
6. Otherwise use the residual distribution of the higher-complexity treatment family.

## ICU and extended stay

NME should increase with projected stay, but only the daily portion should scale.

additional\_day\_NME =  
ward/ICU consumables  
+ F&B  
+ expected underpads/EZ bath  
+ applicable equipment  
+ conditional investigation reserve

Do not multiply fixed medical-record, instrument or laser charges by LOS.

ICU days should be a separate model feature because HME filters, respiratory masks, underpads, warmers and equipment use are materially different from ordinary ward days.

## Required database contracts

Create:

- `fc_curated.nme_item_rules`
- `fc_curated.nme_item_aliases`
- `fc_curated.treatment_nme_profiles`
- `fc_curated.nme_profile_builds`
- `fc_clean.v_nme_item_rules`
- `fc_clean.v_treatment_nme_profiles`
- `fc_clean.v_nme_readiness`
- `fc_estimate.nme_calculations`

Each item rule should contain:

- Service/pharmacy code.
- Canonical NME category.
- Payer applicability.
- Package inclusion condition.
- Patient-payable flag.
- DNB flag.
- Unit: fixed, per day, per use or conditional.
- Current tariff source.
- Effective dates.
- Rule source and confidence.

Each historical profile should contain:

- Treatment/combo/family.
- Payer and organization.
- Package/open status.
- LOS band and ICU context.
- Admission count.
- Positive-NME probability.
- P25/P50/P75/P80/P90.
- Data-quality exclusions.
- Build version and effective period.

## Data-quality controls

Before fitting profiles:

- Exclude negative NME amounts—there are 27 such insurance records.
- Exclude or cap extreme observations above the treatment-level P99.
- Ensure NME does not exceed the final bill.
- Use **HIMS NME Amount**, not the counsellor's NME estimate, as the historical outcome.
- Keep package inclusions and patient-payable NME separate from insurer-payable exclusions.
- Do not train Cash and insurance together.
- Do not use the sparse Corporate history as an automatic model.

## Recommended initial implementation

Phase 1 can immediately produce useful ranges using:

- Final HIMS NME totals.
- Canonical treatment.
- GIPSA/Non-GIPSA.
- Package/open.
- LOS band.
- ICU presence.
- Treatment-family fallback.

Phase 2 should add:

- Code-level NME classification.
- Routine/day-based reconstruction.
- Treatment-specific equipment and consumable profiles.
- Package inclusion/exclusion overrides.
- Organization/MOU rules.
- Residual modelling.

This will give the patient a realistic ballpark while remaining explainable:

Estimated NME payable by patient: ₹6,000–₹10,000

Includes:

- Routine non-medical consumables
- Medical records and food-related charges
- Expected treatment-specific disposables

May increase if:

- Additional equipment/laser is required
- Stay extends beyond the planned LOS
- Conditional investigations are not approved

The critical design decision is to present NME as a separate patient-payable range, not hide it inside the total estimate and not confuse it with package exclusions, copay or DNB items.